Experiment:-18

Problem Statement:-

In many real-world applications such as log analysis, report generation, and glossary processing, it is often required to access the last few entries of a file and display them in reverse order. Bash scripting provides support for arrays, which can be effectively used to store file content and manipulate it programmatically.

The problem is to write a shell script that reads the last few lines of a glossary file, stores them in an array, and prints them in reverse order, demonstrating the use of arrays and reverse traversal in Bash scripting.

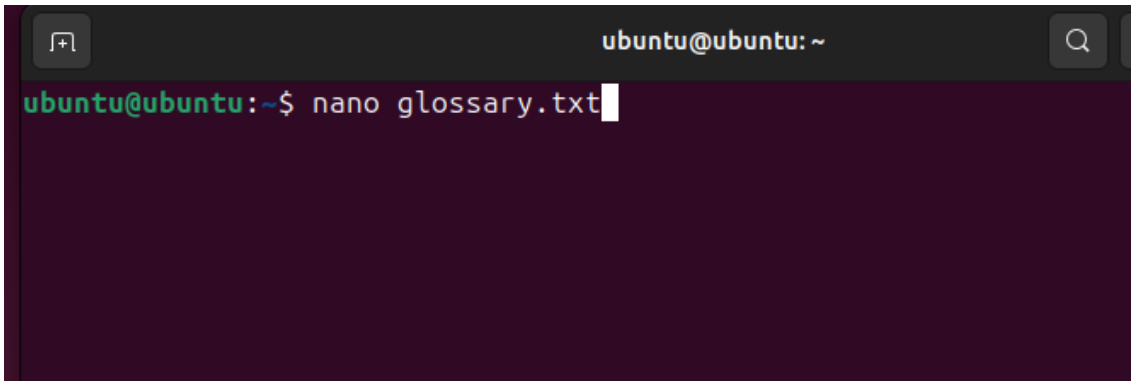
Aim:-Understanding virtual terminal, display manager, X client and window managers in linux

Procedure:-

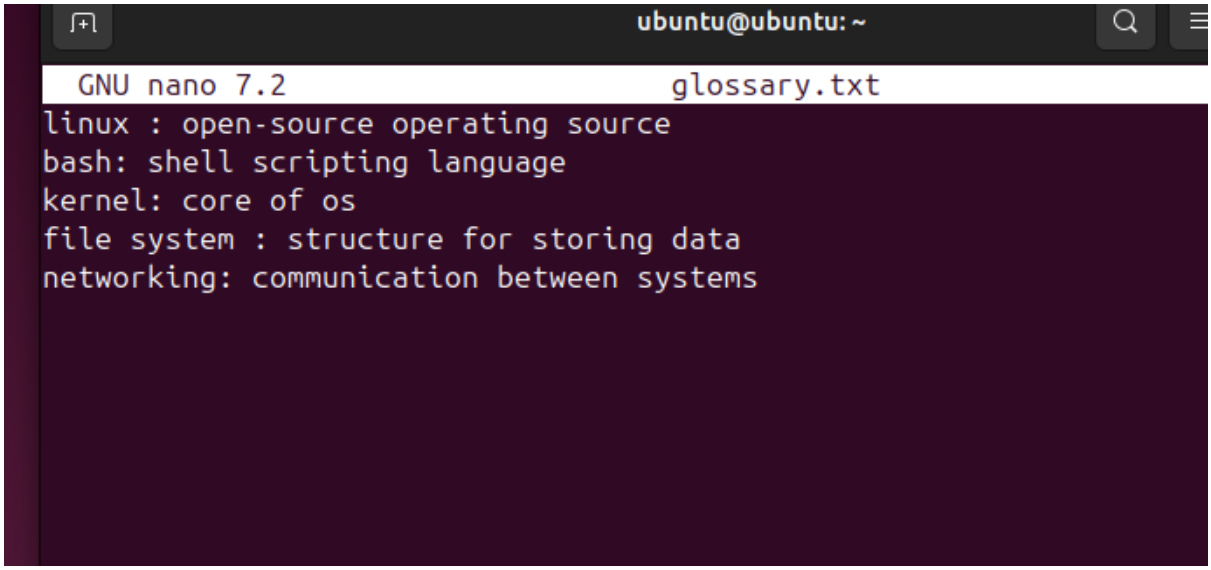
Procedure A. Creating a Glossary File

1. Open the terminal.

2. Create a glossary file: nano glossary.txt



3. Enter sample content: Linux: Open-source operating system , Bash: Shell scripting language, Kernel: Core of operating system, File System: Structure for storing data , Networking: Communication between systems

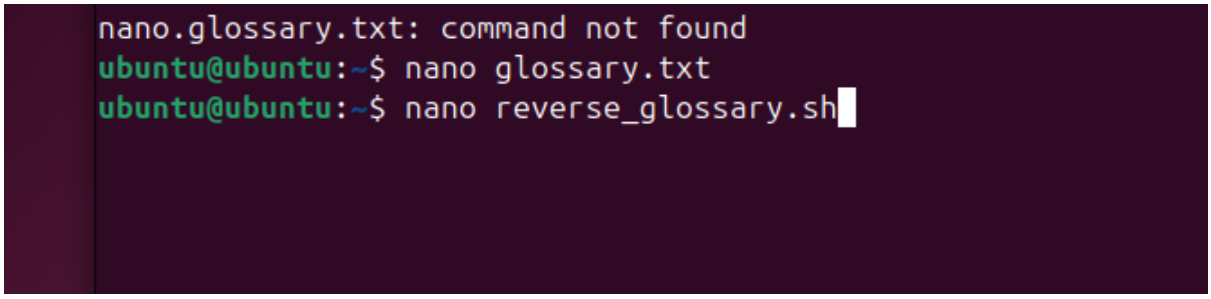


4. Save and exit the editor.

B. Writing the Shell Script

5. Create a script file:

nano reverse_glossary.sh



6. Enter the following script:

```
ubuntu@ubuntu: ~
GNU nano 7.2                    reverse_glossary.sh *
#!/bin/ bash
declare -a lines
index=0
while read line
do
    lines[$index} = "$line"
    index =$((index +1))
done < <(tall -n 5 glossary.txt)
echo "glossary entries in reverse order:"
for ((i=index-1; i>=0;i--))
do
    echo "${lines[$i]}"
done
```

7. Save and exit the editor.

8. Make the script executable: chmod +x reverse_glossary.sh

```
ubuntu@ubuntu:~$ nano reverse_glossary.sh
ubuntu@ubuntu:~$ chmod +x reverse_glossary.sh
```

C. Executing the Script

9. Run the script: ./reverse_glossary.sh

```
└─$ ./reverse_glossary.sh
glossary entries in reverse order:
networking: communication between systems
file system: structure for storing data
kernel: core of operating system
bash: shell scripting language
linux: open-source operating syatem
```